CH01 - Basic & Trigonometric Signals

Common signals

In signal processing, these four signals—**Impulse, Step, Ramp, and Sine**—are considered "standard test signals" because they are the simplest mathematical models for the most common real-world physical forces.

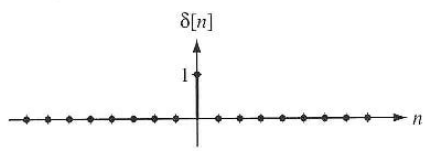
They act as "stress tests" for a system, allowing engineers to predict how a device will handle sudden shocks, steady changes, or constant vibrations without having to test every possible scenario

Unit Impulse Signal

The Unit Impulse models a **sudden, instantaneous impact** or "shock" to a system, like a hammer hitting a bell or a lightning strike on a power line.

Commonly called the ***Dirac Delta function***, this is a theoretical signal that is infinitely high and thin but has an area of exactly **1**.

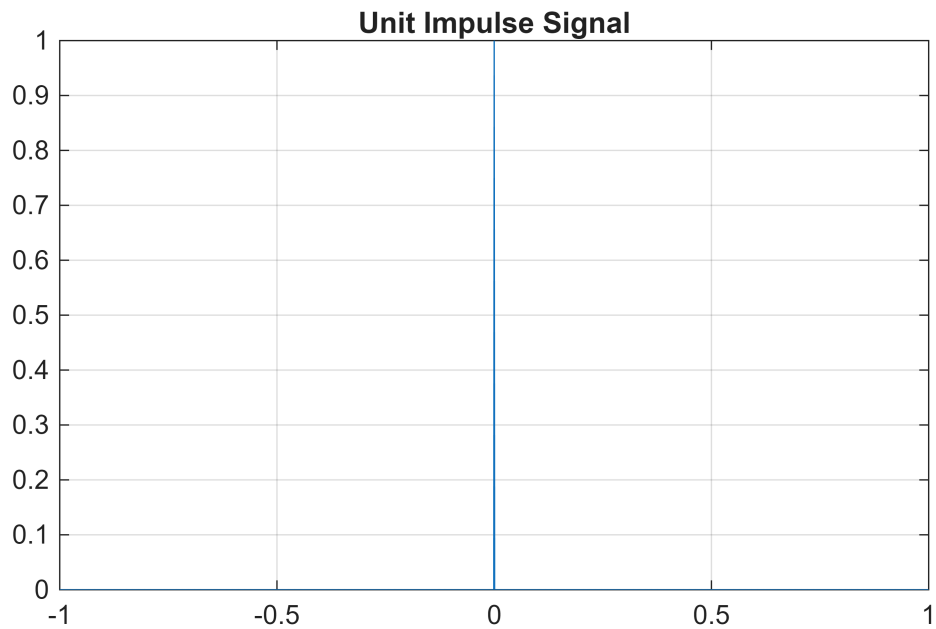
$$\boxed{\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}}$$



Behavior: It exists only at **t = 0** and is zero everywhere else.

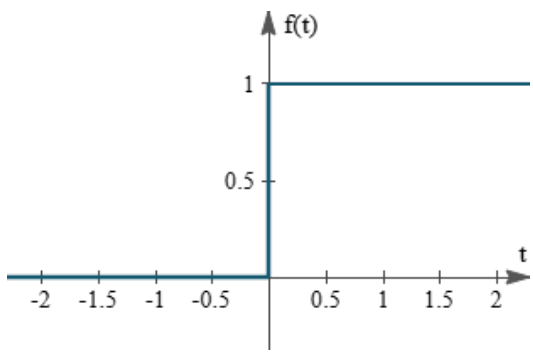
Purpose: It reveals the "Impulse Response," which is the "DNA" of a system. If you know how a system reacts to an impulse, you can mathematically predict how it will react to *any* other input through a process called convolution.

```
t = -1:0.001:1;
delta = (t == 0);

plot(t, delta);

title('Unit Impulse Signal');
grid on;
```



Unit Step Signal

The Unit Step models a **sudden change that stays constant**, like flipping a light switch or a thermostat turning on a heater.



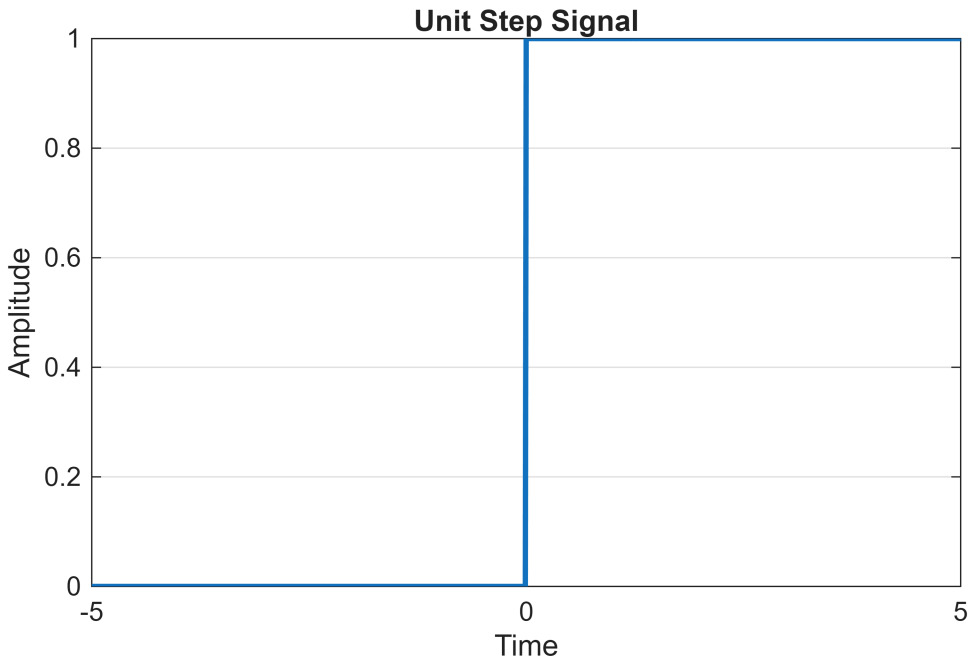
Behavior: It is 0 for all time before $t = 0$, and jumps to exactly **1** for all time after $t = 0$.

Purpose: It is used to test the **stability** and settling time of a system (how long it takes to react to a sudden change).

It is used to analyze **Transient Response**. It shows how much a system "overshoots" its target and how long it takes to settle down (stability).

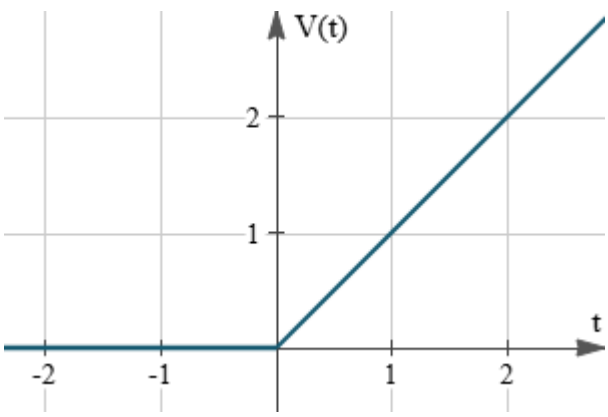
```
t = -5:0.01:5;
u = t >= 0;

plot(t, u, 'LineWidth', 2);
grid on;
title('Unit Step Signal');
xlabel('Time');
```

```
ylabel('Amplitude');
```



Unit Ramp Signal

A ramp signal represents a constant change over time, like an airplane steadily gaining altitude.

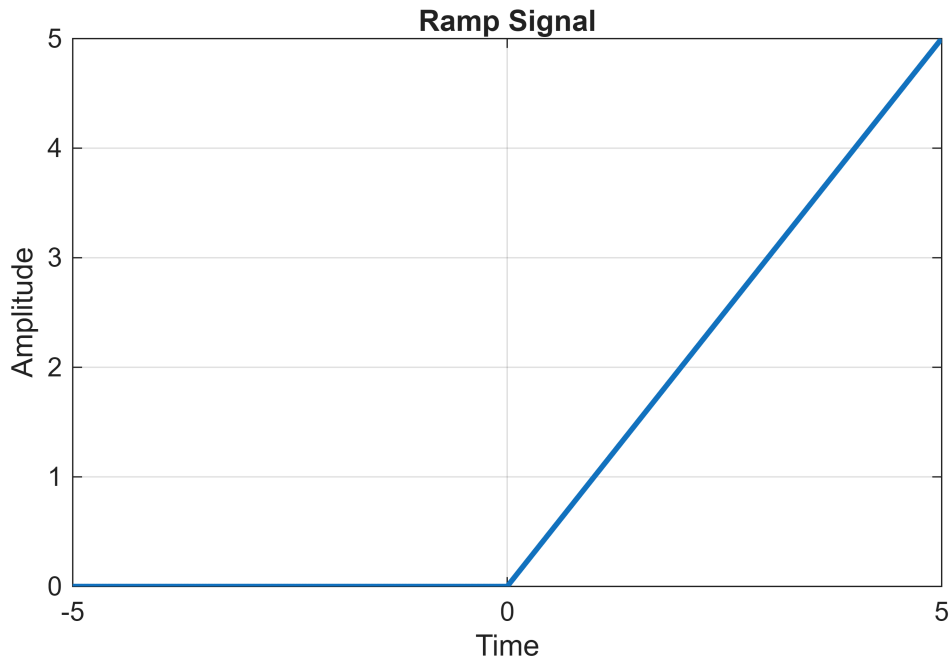


Behavior: It is 0 before t = 0 , and then increases linearly forever.

Purpose: It tests the system's ability to track a moving target. It helps engineers find "steady-state error"—basically, how far behind the system falls when trying to keep up with a constant change.

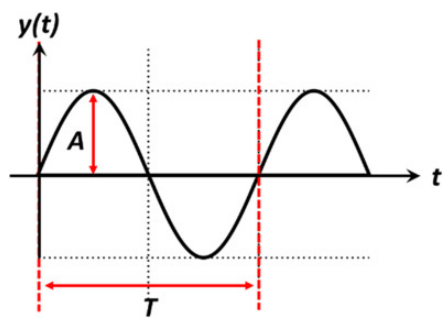
```
t = -5:0.01:5;
r = t .* (t >= 0);

plot(t, r, 'LineWidth', 2);
grid on;
title('Ramp Signal');
xlabel('Time');
```

```
ylabel('Amplitude');
```



Sinusoidal Signal

The most common real-world signal, found in everything from sound waves to the AC power in your walls.



Behavior: It oscillates smoothly and periodically between positive and negative peaks.

Purpose: It is the foundation of **Frequency Analysis** (Fourier Transforms). Every complex signal can be broken down into a collection of sine waves.

```
t = 0:0.1:9*3.14
```

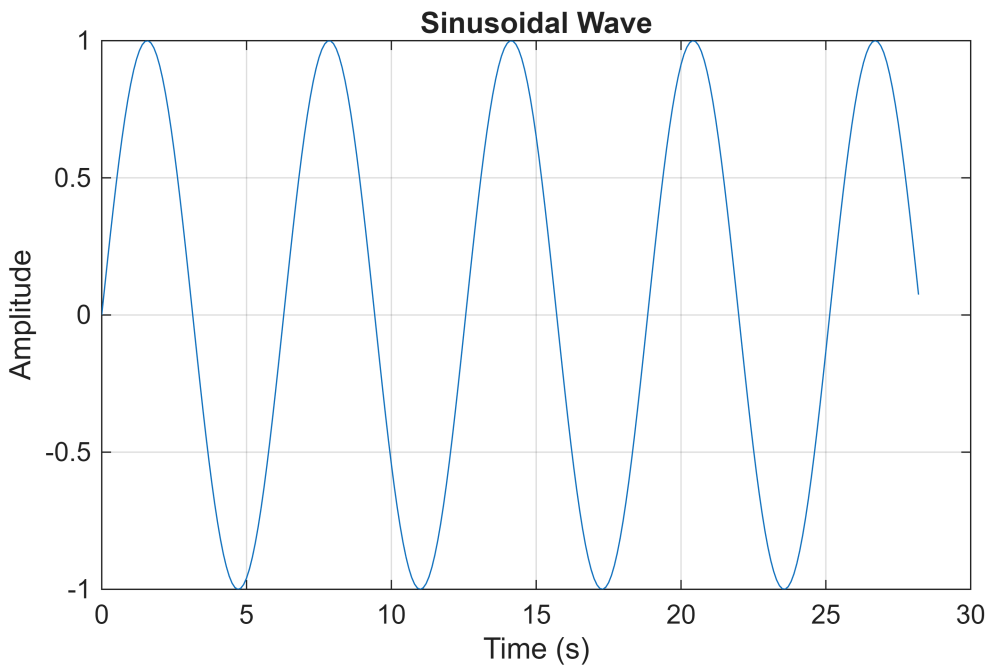
```
t = 1×283
         0    0.1000    0.2000    0.3000    0.4000    0.5000    0.6000    0.7000 ···
```

```
y = sin(t)
```

```
y = 1×283
         0    0.0998    0.1987    0.2955    0.3894    0.4794    0.5646    0.6442 ···
```

```
plot(t,y)
xlabel('Time (s)');
```

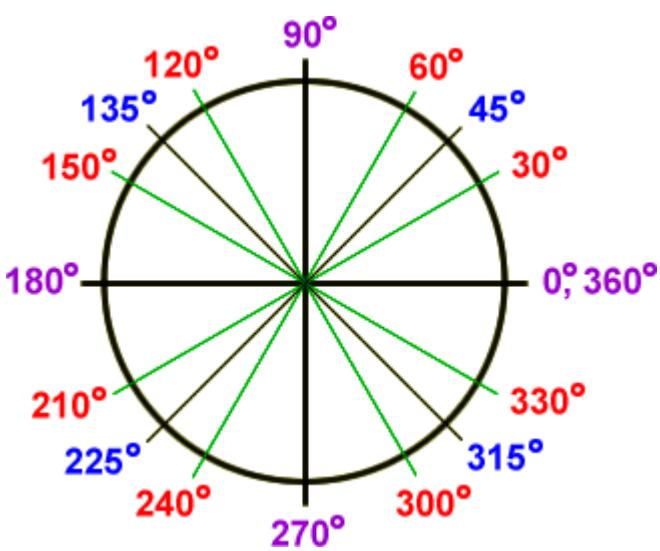
```
ylabel('Amplitude');
title('Sinusoidal Wave');
grid on;
```



Basic Trigonometric Signals

1. **Degrees**

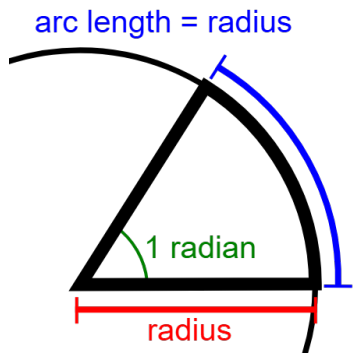
Degrees are based on tradition. We decided a full circle is 360°. Why 360? It's a "friendly" number—highly divisible by 2, 3, 4, 5, 6, 8, 9, 10, and 12.



Mental Image: Think of a compass or a clock. A right angle is 90°, and a straight line is 180°.

2. **Radians**

Radians are based on the actual geometry of the circle. **One radian is the angle created when you take the radius of a circle and wrap it around the edge (the arc)**.



The Math: Since the circumference of a circle is 2*pi*r, there are exactly 2*pi radians in a full circle.

Mental Image: If you walked a distance equal to the radius along the curve of a circle, the angle you've covered is exactly 1 radian.

The Shortcut

- **Radians to Degrees:** Multiply by $\frac{180}{\pi}$
- **Degrees to Radians:** Multiply by $\frac{\pi}{180}$

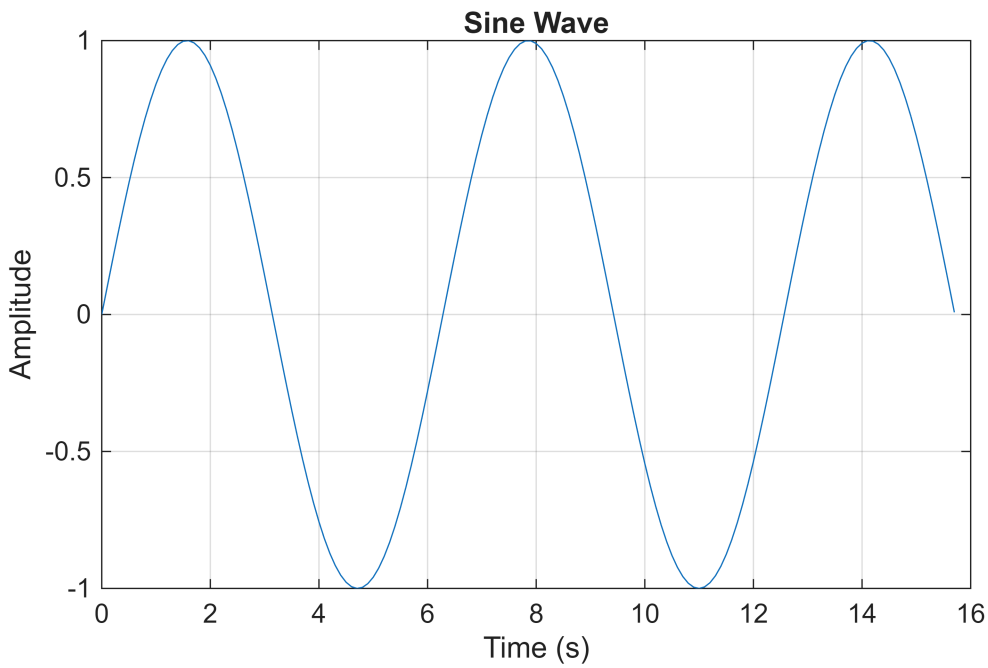
```
t = 0:0.1:5*3.14
```

```
t = 1×158
         0    0.1000    0.2000    0.3000    0.4000    0.5000    0.6000    0.7000 ···
```

```
y = sin(t)
```

```
y = 1×158
         0    0.0998    0.1987    0.2955    0.3894    0.4794    0.5646    0.6442 ···
```

```
plot(t,y)
xlabel('Time (s)');
ylabel('Amplitude');
title('Sine Wave');
grid on;
```



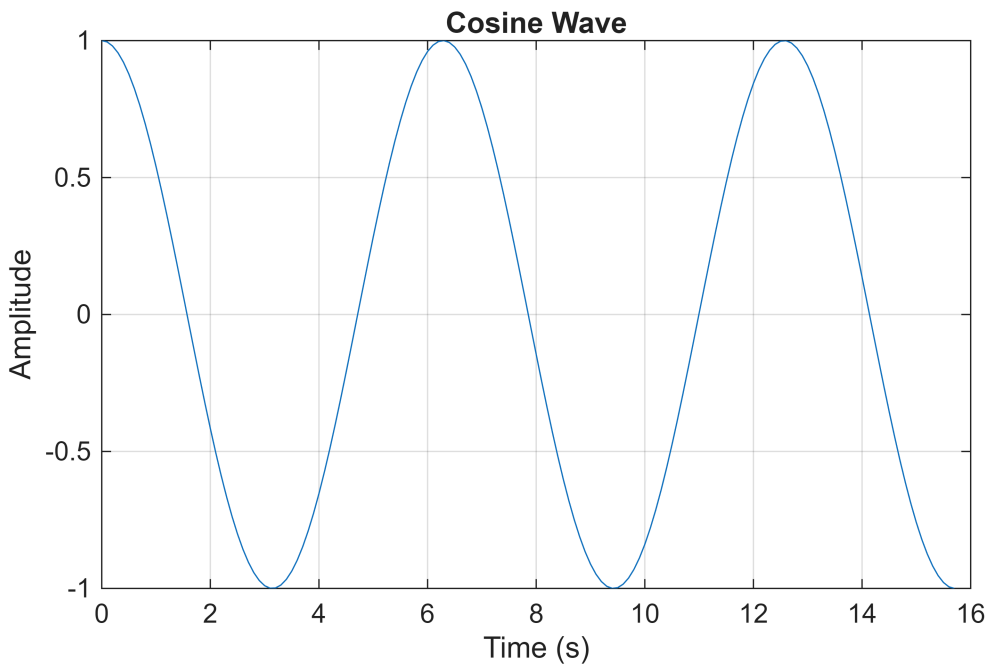

```
t = 0:0.1:5*3.14
```

```
t = 1×158
         0    0.1000    0.2000    0.3000    0.4000    0.5000    0.6000    0.7000 ···
```

```
y = cos(t)
```

```
y = 1×158
    1.0000    0.9950    0.9801    0.9553    0.9211    0.8776    0.8253    0.7648 ···
```

```
plot(t,y)
xlabel('Time (s)');
ylabel('Amplitude');
title('Cosine Wave');
grid on;
```



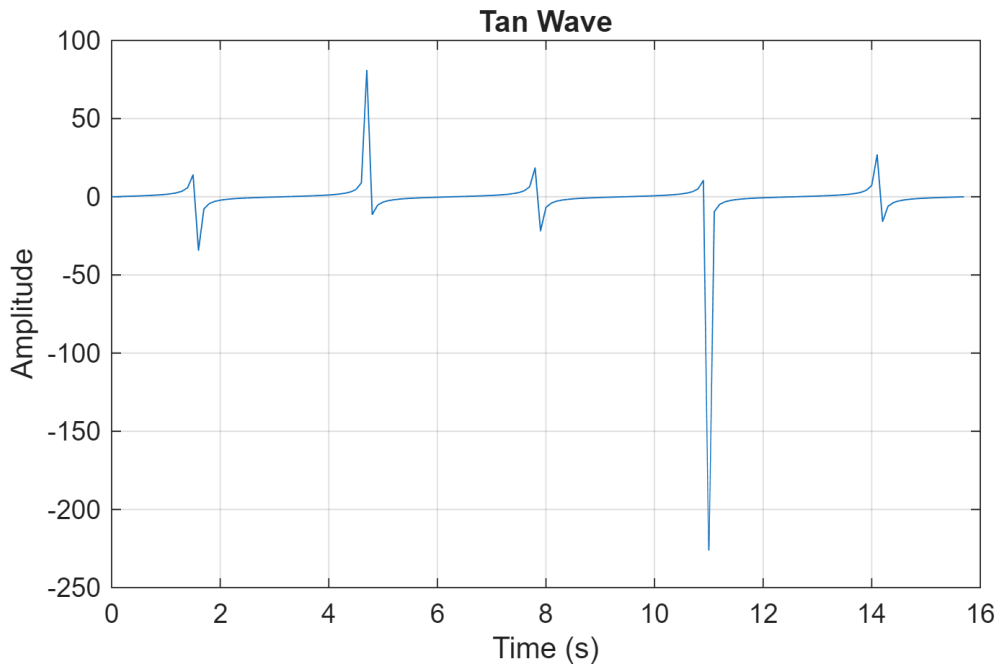

```
t = 0:0.1:5*3.14
```

```
t = 1×158
         0    0.1000    0.2000    0.3000    0.4000    0.5000    0.6000    0.7000 ···
```

```
y = tan(t)
```

```
y = 1×158
         0    0.1003    0.2027    0.3093    0.4228    0.5463    0.6841    0.8423 ···
```

```
plot(t,y)
xlabel('Time (s)');
ylabel('Amplitude');
title('Tan Wave');
grid on;
```



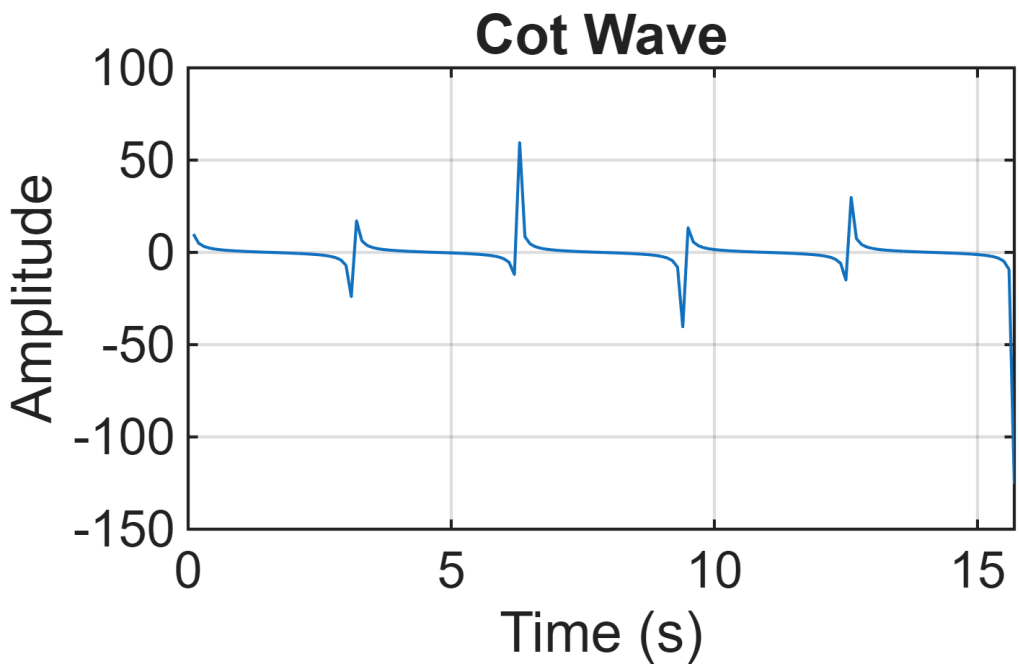

```
t = 0:0.1:5*3.14
```

```
t = 1×158
         0    0.1000    0.2000    0.3000    0.4000    0.5000    0.6000    0.7000 ···
```

```
y = cot(t)
```

```
y = 1×158
       Inf    9.9666    4.9332    3.2327    2.3652    1.8305    1.4617    1.1872 ···
```

```
plot(t,y)
xlabel('Time (s)');
ylabel('Amplitude');
title('Cot Wave');
grid on;
```



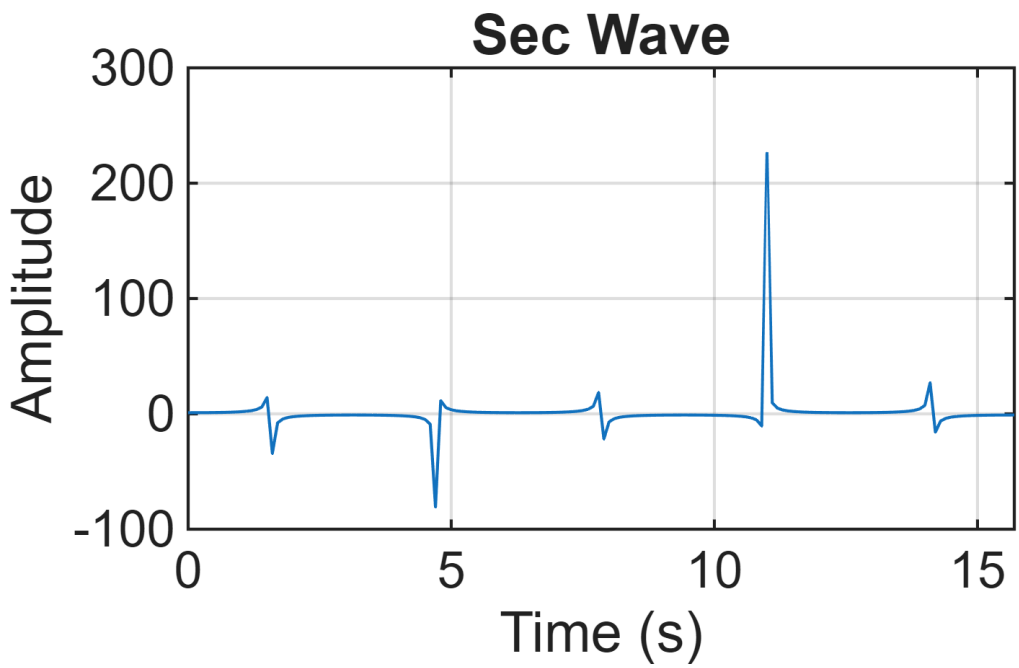

```
t = 0:0.1:5*3.14
```

```
t = 1×158
         0    0.1000    0.2000    0.3000    0.4000    0.5000    0.6000    0.7000 ···
```

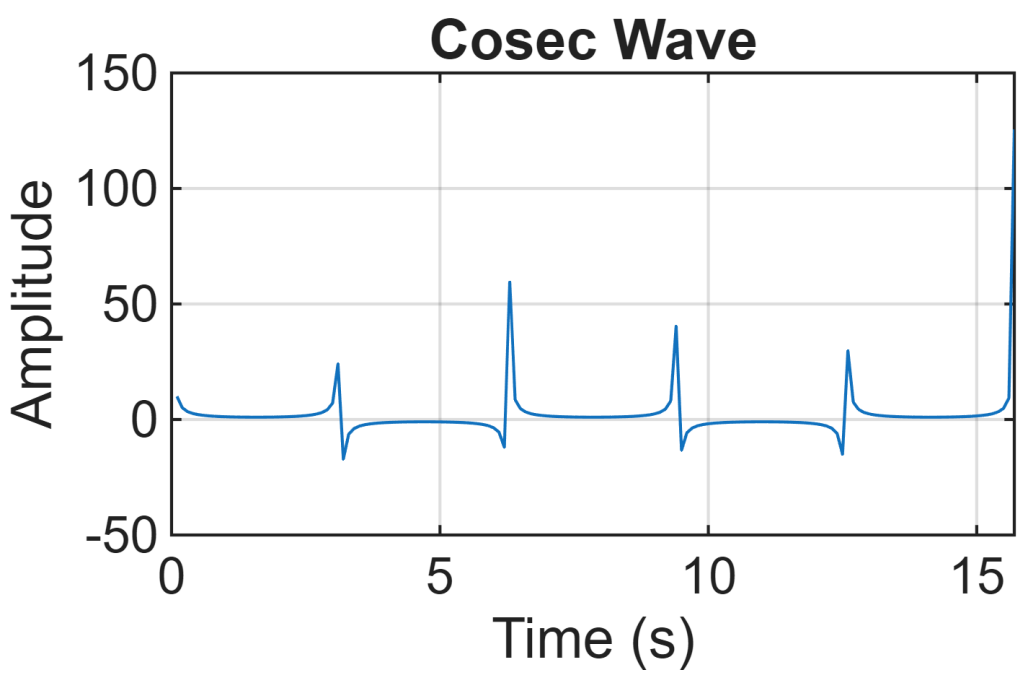
```
y = sec(t)
```

```
y = 1×158
    1.0000    1.0050    1.0203    1.0468    1.0857    1.1395    1.2116    1.3075 ···
```

```
plot(t,y)
xlabel('Time (s)');
ylabel('Amplitude');
title('Sec Wave');
grid on;
```




```
t = 0:0.1:5*3.14
```

```
t = 1×158
         0    0.1000    0.2000    0.3000    0.4000    0.5000    0.6000    0.7000 ···
```

```
y = 1./sin(t);
plot(t,y)
xlabel('Time (s)');
ylabel('Amplitude');
title('Cosec Wave');
grid on;
```